

33 Which statement about the transport of sucrose in plants is correct?

- A** The hydrogen ion (proton) concentration gradient, on which the transport of sucrose into companion cells depends, is maintained by active transport.
- B** Sucrose moves through plasmodesmata by passive diffusion from mesophyll cells into companion cells down the concentration gradient.
- C** ATP synthesised by the mitochondria in companion cells provides the energy for the active transport of sucrose from companion cells into phloem sieve tube elements.
- D** Water enters companion cells by active transport causing hydrostatic pressure to build up and push sucrose into phloem sieve tube elements by mass flow.